

Hugh Everett III

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The difficulties of quantum gravity are both technical—calculations tend to blow up and give infinitely big answers—and also conceptual. Even in quantum mechanics, while you might not be able to say precisely where a certain particle is, the notion of “a point in space” is perfectly well defined. We can specify a location and ask what is the probability of finding the particle nearby. But if reality doesn’t consist of stuff distributed through space, but rather is a quantum wave function describing superpositions of different possible spacetimes, how do we even ask “where” a certain particle is observed?

The puzzles become worse when we turn to the measurement problem. By the 1950s the Copenhagen school was established doctrine, and physicists had made their peace with the idea of wave functions collapsing when a measurement occurred. They were even willing to go along with treating the measurement process as a fundamental part of our best description of nature. Or, at least, not to fret too much about it.

But what happens when the quantum system under consideration is the entire universe? Crucial to the Copenhagen approach is the distinction between the quantum system being measured and the classical observer doing the measuring. If the system is the universe as a whole, we are all inside it; there’s no external observer to whom we can appeal. Years later, Stephen Hawking and others would study quantum cosmology to discuss